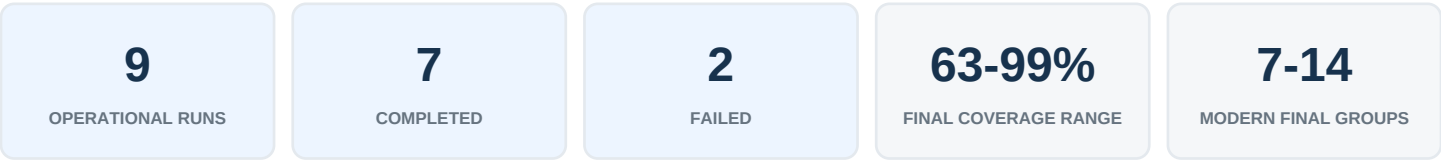


Quantitative analysis of all saved pipeline runs

A descriptive audit of every operational run saved through August 4, 2026. The report separates infrastructure failures, the legacy non-monotonic prototype, fresh runs, fixed-corpus labeling treatments, and controlled assignment-evidence treatments.



Executive findings

1. Assignment evidence forms a breadth gradient. With an identical pre-epoch state, final coverage was 94.7% with questions only, 87.6% with representative paragraphs, and 75.9% with every group paragraph.	2. Full-paper labels changed topology more than coverage. On the same 323 fixed paragraphs, fixed-section and full-paper labels ended at 92.9% and 92.0% coverage, but at 14 versus 7 groups.
3. Representative evidence is the middle condition. It retained 90.0% final-assignment Jaccard overlap with question-only assignment, versus 76.6% for all group evidence, while leaving 40 rather than 78 orphans.	4. These are behavior measurements, not quality scores. Coverage, group count, and overlap show how the pipeline behaves. No run establishes whether assignments or analogies are conceptually correct.

Run inventory

Run	Condition	Status	Min	Paras	Groups	Coverage	Orphans
08/01 171253	Bootstrap failure	failed	0.3	-	-	-	-
08/01 171519	Legacy reassignment	complete	11.3	224	2	63.4%	82
08/03 153516	Monotonic baseline	complete	5.5	314	10	99.4%	2
08/03 190623	Question-only replication	complete	9.0	323	12	94.7%	17
08/03 192347	Full-paper labels	complete	9.9	323	7	92.0%	26
08/03 193353	All group evidence	complete	5.5	323	11	75.9%	78
08/03 210200	Fixed-section labels	complete	12.5	323	14	92.9%	23
08/03 211500	Representative group evidence attempt	failed E3	5.2	323	12*	77.4%*	73*
08/03 212300	Representative group evidence	complete	0.9	323	10	87.6%	40

* Partial values are the last durable E2 snapshot from the failed representative-evidence attempt, not a final outcome. The completed representative run has the same configuration and identical E0-E2 results, so the failed attempt is not an independent replicate.

What each condition means

The condition name is shorthand for which information was held fixed and which information Claude received. The definitions below are part of the analysis, not merely run labels.

Condition	Plain-language definition
Bootstrap failure	Infrastructure launch attempt. It stopped before section extraction because the server used the wrong Python environment, so it contains no analytical outcome.
Legacy reassignment	Earlier non-monotonic prototype. It retained only two initial supergroups and reassigned every paragraph on every epoch, so coverage could rise or fall and its group count is not comparable to the current complete group set.
Monotonic baseline	First complete append-only implementation. It ran fresh extraction and sticky orphan assignment, but predates the controlled fixed-corpus experiment metadata.
Question-only replication	Fresh end-to-end rerun from the three PDFs. During orphan assignment, Claude saw each complete orphan paragraph and the candidate group questions, but no text from paragraphs already assigned to those groups.
Fixed-section labels	Reused the replication's exact paragraph ids, boundaries, text, sections, order, and relations. Claude regenerated only the paragraph questions while seeing each paragraph's complete containing section; downstream matching and epochs were rerun.
Full-paper labels	Reused the same exact fixed paragraph corpus, but Claude regenerated paragraph questions while seeing the complete paper. This tests broader labeling context without changing paragraph boundaries or source text.
Representative group evidence	Copied the question-only replication's complete pre-epoch state. During orphan assignment, each group question was accompanied by deterministic corpus-TF-IDF medoid paragraphs: at most two per source paper and six per group.
All group evidence	Copied the same pre-epoch state, then accompanied every candidate group question with the complete text of every paragraph already assigned to that group. This is what 'all group evidence' means; only assignment evidence changed.

Controlled comparison structure

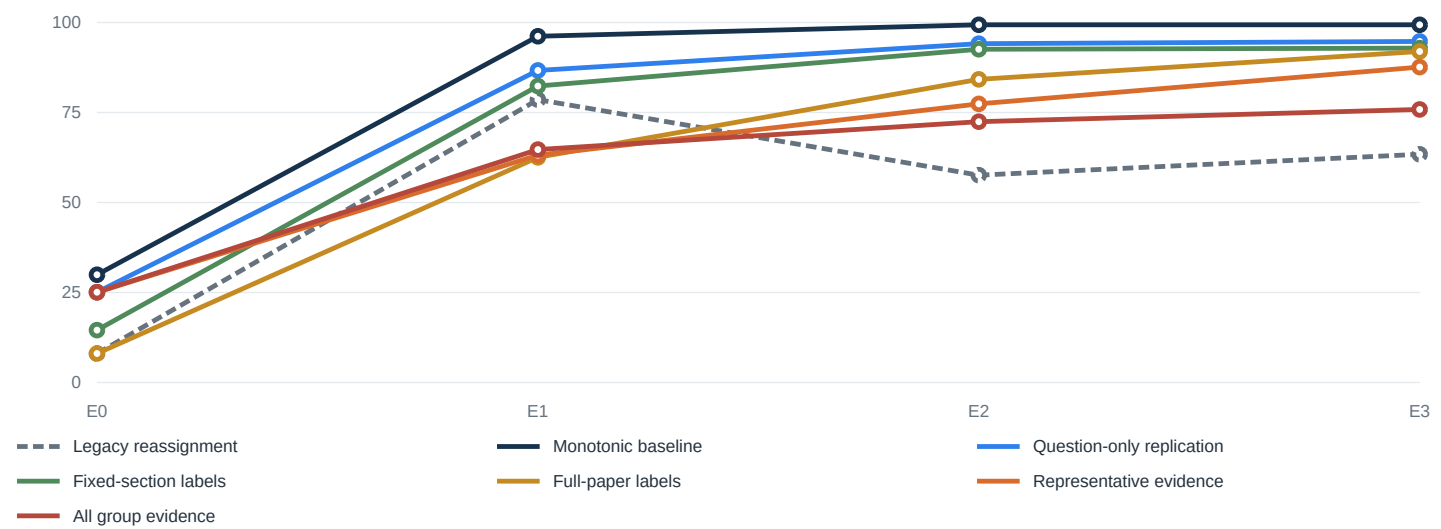
Label-context comparison Fixed-section and full-paper labels reuse identical paragraph ids, text, order, sections, and relations. Both use question-only orphan assignment. Only the context used to regenerate paragraph questions differs.	Assignment-evidence comparison Question-only, representative evidence, and all group evidence share the exact same pre-epoch files. The assignment prompt's view of existing group members is the controlled difference.
Fresh-run comparison The monotonic baseline and question-only replication reran extraction from PDFs. Their paragraph corpora differ, so they measure end-to-end instability rather than a clean prompt ablation.	Failure accounting Both failed runs remain in the inventory. A failed run counts operationally but contributes no final treatment outcome; a durable partial epoch is marked explicitly.

How to read the metrics

Coverage is the percentage of extracted paragraphs assigned to at least one group. **Memberships** count every many-to-many placement. **Multi** counts paragraphs present in more than one group. **Orphans** are paragraphs assigned to no group. Fewer groups indicate broader aggregation, not necessarily a better abstraction.

Coverage across epochs

E0 is the initial state; E1-E3 are refinement epochs. Current append-only runs can only gain coverage. The dashed legacy series can lose coverage because it reassigned every paragraph at each epoch.



Run dynamics

Condition	Coverage path (%)	Group-count path	Net unique assigned	Final memberships	Multi
Legacy reassignment	8.0 / 78.6 / 57.6 / 63.4	2 / 2 / 2 / 2	+158 / -47 / +13	204	62
Monotonic baseline	29.9 / 96.2 / 99.4 / 99.4	19 / 12 / 10 / 10	+208 / +10 / +0	328	16
Question-only replication	25.1 / 86.7 / 94.1 / 94.7	19 / 14 / 12 / 12	+199 / +24 / +2	339	31
Fixed-section labels	14.6 / 82.4 / 92.6 / 92.9	19 / 16 / 14 / 14	+219 / +33 / +1	305	5
Full-paper labels	8.0 / 62.5 / 84.2 / 92.0	10 / 8 / 7 / 7	+176 / +70 / +25	309	12
Representative evidence	25.1 / 63.2 / 77.4 / 87.6	19 / 14 / 12 / 10	+123 / +46 / +33	283	0
All group evidence	25.1 / 64.7 / 72.4 / 75.9	19 / 14 / 12 / 11	+128 / +25 / +11	245	0

Most assignments occurred in E1, but later epochs mattered for the context-heavy conditions. Full-paper labels added 176, 70, and 25 unique paragraphs across E1-E3; representative evidence added 123, 46, and 33. The fixed-section condition nearly saturated by E2, adding only one paragraph in E3.

Controlled comparisons

A. Assignment evidence: what existing group content does Claude see?

All three runs copy the question-only replication's pre-epoch artifacts byte-for-byte. The orphan paragraph is always supplied in full; the treatment changes only the evidence attached to candidate group questions.

Condition	Existing group-member text supplied	Evidence chars E1/E2/E3	Coverage	Orphans	Groups	Multi	Jaccard vs Q-only
Question-only replication	None; group question only	0k / 0k / 0k	94.7%	17	12	31	100.0%
Representative group evidence	TF-IDF medoids; max 2/paper, 6/group	36k / 45k / 39k	87.6%	40	10	0	90.0%
All group evidence	Every paragraph already assigned to group	59k / 123k / 137k	75.9%	78	11	0	76.6%

Coverage declines from 94.7% to 87.6% to 75.9% as existing-member evidence becomes richer. Representative evidence preserves 279 of the question-only run's final assignments and adds 4 not present there; all group evidence preserves 239 and adds 6. Both evidence-aware treatments produced zero many-to-many assignments. This pattern is consistent with more conservative assignment, but one run per treatment cannot identify why.

B. Paragraph-question context: section versus complete paper

The strict comparison is fixed-section versus full-paper: both preserve the same 323 paragraph units and rerun downstream stages with question-only assignment.

Condition	Question-label context	Labels changed	Mean question words	E0 groups / coverage	Final groups / coverage	Orphans
Question-only replication	Fresh section extraction; boundaries and labels regenerated	not fixed	9.2	19 / 25.1%	12 / 94.7%	17
Fixed-section labels	Fixed units; complete containing section	322/323	15.1	19 / 14.6%	14 / 92.9%	23
Full-paper labels	Fixed units; complete paper	323/323	16.4	10 / 8.0%	7 / 92.0%	26

Fixed-section and full-paper labels end with nearly the same coverage (92.9% versus 92.0%), yet full-paper context halves the final group count from 14 to 7. It also produces longer questions on average (16.4 versus 15.1 words). This supports a strong context effect on clustering topology, not a claim that broader clusters are more correct.

C. Reproducibility and failed attempts

The monotonic baseline extracted 314 paragraphs and ended at 99.4% coverage in 10 groups. The fresh question-only replication extracted 323 and ended at 94.7% in 12 groups. Because extraction and labels were regenerated and the harness changed, this is end-to-end instability rather than a clean repeatability estimate. The failed representative-evidence attempt reached E2 at 77.4% coverage before `refine_epochs.py` exited non-zero; its completed replacement reused identical cached E0-E2 outputs, so it is recovery evidence, not an independent replicate.

What the current runs support

Supported Append-only assignment makes coverage monotonic. Labeling context materially changes group topology. Supplying existing group paragraphs suppresses assignment breadth, with representative evidence intermediate between question-only and all-evidence prompts.	Not established Higher coverage is not higher accuracy. Fewer groups are not necessarily better analogies. One run per controlled treatment cannot separate prompt effects from model stochasticity or cache-path effects.
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Recommended next technical evaluation

Freeze one paragraph corpus and one pre-epoch state, clear treatment-specific caches, and run at least five independent replicates of question-only, representative-evidence, and all-group-evidence assignment. Human-label a stratified sample of assigned and orphan paragraphs. Report assignment precision and recall, top-1 agreement, orphan correctness, group coherence, analogical-role agreement, and between-run stability alongside coverage and group count.